

Face, eyes and teeth

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Having looked at it on YouTube, I don't like to look at it too much because it freaks me out a bit. The ball hit me in the eye and went two centimetres back.

South African cricket player Mark Boucher on his career-ending eye injury

Injuries to the face in sport usually result from direct trauma. After reviewing clinical assessment and soft tissue injury management, we outline management of injuries to the nose, ear, eye, teeth and facial bones.

FUNCTIONAL ANATOMY

The bones of the face are shown in Figure 22.1. As most of these bones are subcutaneous, they are easily examined. Examination should include palpation of the forehead and supraorbital rims for irregularities and contour deformities.

The orbit is a cone-shaped cavity, with a margin that consists of the supraorbital ridge above, the infraorbital margin below, the zygomatic arch laterally and the nasal bone medially. The recess formed protects the eye from a blow from a large object. A smaller, deformable object such as a squash ball may, nevertheless, compress the globe and orbital contents, leading to a 'blow-out' fracture in the orbital floor that orbital content can herniate through.

The zygomatic arch of the malar bone creates the prominence of the cheek. Fractures in this region may cause flattening of the cheek and a palpable irregularity in the inferior orbital margin.

The maxilla forms the upper jaw. Its superior surface helps create the floor of the orbit and the inferior surface forms the major part of the hard palate. Mobility of the hard palate, determined by grasping the central incisors, may indicate a maxillary fracture.

The lower jaw consists of the horseshoe-shaped mandible. The mandible is made up of body, angle and ramus, which are easily palpated. The coronoid process can be palpated by a direct intraoral approach. The gingiva overlying the alveolar ridge may be lacerated in mandibular body fractures.

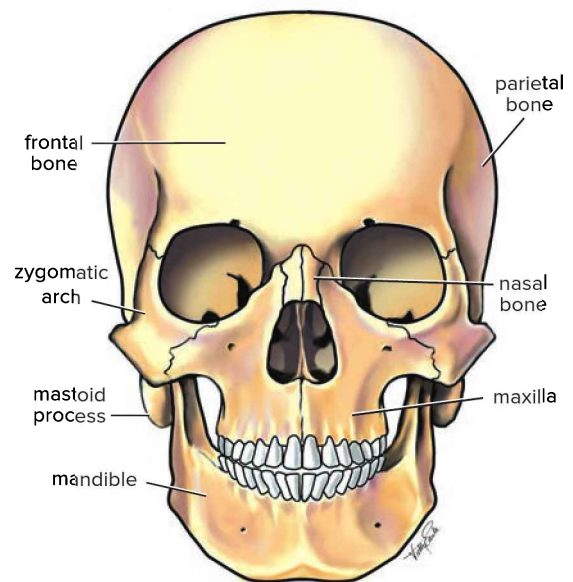


Figure 22.1 Facial bones

CLINICAL ASSESSMENT

Facial injuries¹ are frequently associated with profuse bleeding. While it is important to control the bleeding, it is also vital to fully assess the underlying structures. All head and neck injuries should be considered closed head injuries. Cervical spine precautions should be taken if the patient is unconscious, has neurological deficits or cervical spine tenderness, or if cervical spine injury is feasible considering the mechanism of injury. The airway is particularly vulnerable to obstruction because

of bleeding, structural compromise of bony structures (e.g. mandible), or dislodged teeth, tooth fragments or dental appliances. The practical steps to assess facial injuries are:

- ascertain the mechanism of injury and locate the source of the patient's pain
- check for blurred vision, diplopia (double vision), concussion or cerebrospinal fluid leakage
- inspect the nasal septum and external ear for haematomas and nasal obstruction
- observe facial asymmetry or structural depressions
- look for a sunken eye globe suggestive of a blow-out fracture
- observe lacerations or deep abrasions overlying suspected fractures.

Palpate facial bones (orbital rims, nasal bones, temporo-mandibular joints) for significant tenderness, crepitus, numbness or contour irregularities. Midface instability or crepitus may be demonstrated by stabilising the forehead with one hand while gently pulling on the maxillary incisors with the other gloved hand. Bimanual palpation along the mandible and maxilla (one gloved hand palpating intraorally) will uncover instability, irregularity or tenderness.

Assess extraocular eye movements and cranial nerves III, IV and VI by having the patient keep his or her chin in a fixed position while tracking the examiner's finger movements in all four quadrants. If the patient is able to track the movements without reporting diplopia, acute extraocular nerve entrapment caused by an orbital blow-out fracture can be ruled out. An inability to raise the eyebrow or wrinkle the forehead following laceration to the eyebrow suggests injury to the temporal branch of the facial nerve on that side. Reduced sensation over the skin below the eye in the distribution of the infraorbital nerve may be associated with a blow-out fracture of the orbit. The nerve distribution includes the upper gum and lip.

PRACTICE PEARL

If the patient is unable to open his or her mouth or exhibits severe pain along the lateral aspect of the cheek or jaw when attempting to open, a fracture of the mandible or zygoma must be considered.

With the mouth open, the oral cavity should be assessed to rule out damage to the teeth and lacerations in the intraoral mucosa or tongue. Locate fractured or missing teeth, when possible, to avoid accidental aspiration. When asked to close the mouth, the patient's sense of malocclusion suggests a significant fracture of the mandible, maxilla or palate.

Leakage of cerebrospinal fluid (CSF) following a blow to the nose (CSF rhinorrhea) may indicate a fracture of the base of the anterior cranial fossa. CSF is a clear discharge and

the patient may report a salty taste in the mouth. If there is doubt about the origin of a nasal discharge associated with trauma, the discharge should be tested with a urinary dipstick for glucose. CSF is positive for glucose.

A list of common conditions and conditions not to be missed is shown in Table 22.1.

LACERATIONS AND CONTUSIONS

Lacerations and contusions to the face and scalp are prevalent in sports such as football, ice hockey, martial arts and racquet sports.¹ Examination should include palpation of the underlying bone to detect bony tenderness. Neurological examination is required if there is a history of loss of consciousness or suspected skull fracture.

Immediate management of lacerations

Begin immediate management with ice and pressure to reduce local swelling. Control bleeding with direct pressure over the wound using sterile gauze. A player with a bleeding wound must be removed from the field of play immediately as there is concern that the presence of blood may increase the risk of hepatitis B or human immunodeficiency virus (HIV) infection for other players.

After removing the athlete from the field of play, examine the laceration closely under good light. Further cleaning and removal of foreign bodies may be required. If necessary, infiltrate a local anaesthetic agent to adequately clean the wound. The local anaesthetic used should be 1% or 2% lignocaine (lidocaine) containing adrenalin (epinephrine) 1:100 000 to provide some vasoconstriction as well as analgesia.

Management of larger lacerations

Lacerations greater than 0.25–0.5 cm long should be closed if they appear clean. Closure may be obtained by suturing or by taping with adhesive strips (Steristrips®). Steristrips are ideal for small wounds; however, persistent bleeding or excessive sweating may prevent adhesion. To overcome this, tincture of benzoin (friar's balsam) may be applied to increase adhesiveness. Adequate dressings will be required to keep the adhesive strips in place, especially if the player is returning to the field. Scalp wounds often bleed profusely. Small wounds can be controlled with local pressure, but larger ones require suturing.

If facial lacerations require suturing, use 5/0 or 6/0 nylon. It is important that the skin edges are healthy. Pieces of devitalised skin should be debrided. Take care to approximate the skin edges carefully while suturing. Remove sutures after 5 days and place adhesive strips over the wound for a further week. The wound should be kept dry for at least 48 hours.

An alternative to suturing is skin staples. These must be covered if the player is returning to the field to prevent them from being accidentally torn out or from injuring

Table 22.1 Facial injuries in sport

Category	Common	Less common	Not to be missed
Facial soft tissue	Contusion Laceration		Associated underlying fracture/nerve injury
Nose	Fracture of nasal bones Epistaxis	Fracture of nasal septum	Septal haematoma
Ear	Contusion ('cauliflower ear') Otitis media Otitis externa	Laceration Ruptured tympanic membrane	Fractured petrous temporal bone Torn auditory nerve
Eye	Corneal abrasion Corneal foreign body Conjunctival foreign body Subconjunctival haemorrhage Eyelid laceration	Chemical burns Vitreous haemorrhage Retinal haemorrhage Retinal oedema Hyphaema	Corneal laceration Retinal detachment Lens dislocation Blow-out fracture of the orbit Optic nerve injury Injury to lacrimal system
Teeth	Enamel chip fracture Luxated tooth Avulsed tooth	Crown fracture	Impacted tooth
Facial bones	Temporomandibular joint sprain or malalignment Zygoma fracture Orbital blow-out fracture	Fractured maxilla Fractured mandible	

Table 22.2 Management of specific lacerations

Laceration type	Management considerations
Deep lacerations (e.g. to galea aponeurotica in forehead)	Interrupted 5/0 absorbable sutures (prior to skin closure)
Eyebrow laceration	Do not shave eyebrow hair. Strict anatomical approximation required
Partial-thickness lip laceration at vermilion border	Place first suture at mucocutaneous junction to ensure accurate alignment
Full-thickness lip laceration	Three layer closure required (preferably by plastic surgeon) consisting of: oral mucosa first; orbicularis oris second; skin last
Deep intraoral laceration	Use 3/0 silk sutures and keep in place for 1 week

another player in a collision. Another alternative is the use of histoacryl glue.

Additional considerations

All patients with potentially contaminated wounds should receive tetanus prophylaxis and a short course of oral antibiotic therapy, for example, cephalexin (250–500 mg, 6 hourly) or flucloxacillin (250–500 mg, 6 hourly). Wounds that may be contaminated by another player's saliva, such as bite wounds, should not be closed but should be cleaned meticulously. The player should be treated with oral metronidazole (400 mg, 8 hourly) in addition to penicillin and observed very closely for the development of cellulitis. If signs of infection appear, treat with intravenous antibiotics.

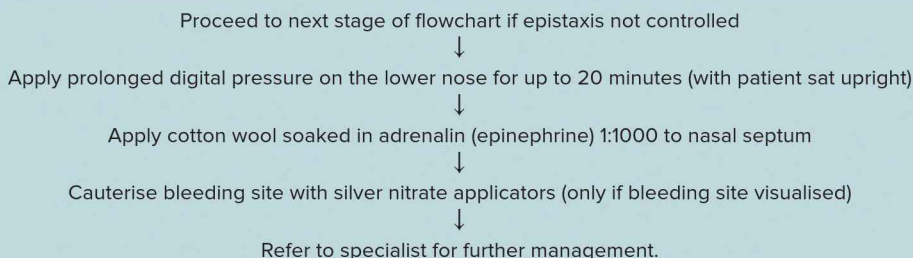
NOSE

The three main conditions that affect the nose in sports medicine are epistaxis (bleeds), fractures and septal haematomas.

Epistaxis (nosebleed)

Nasal haemorrhage occurs frequently in association with nasal injuries. It usually arises from the nasal septum, which receives its blood supply from branches of the internal and external carotid arteries. In most non-traumatic cases, the bleeding arises from a rich plexus of vessels in the anterior part of the septum, known as Little's or Kiesselbach's area (Fig. 22.2), whilst bleeding will originate from the vessels local to the affected area in traumatic cases.

Epistaxis 'game-day' management flowchart



Initial management consists of prolonged direct digital pressure on the lower nose for up to 20 minutes, compressing the vessels on the nasal septum with the patient sitting upright. Cold compresses over the bridge of the nose promote vasoconstriction. If bleeding continues, apply cotton wool soaked in adrenalin (epinephrine) 1:1000 to the nasal septum. If the bleeding site can be located, it may be cauterised with silver nitrate applicators (cotton swabs soaked in 4% trichloroacetic acid).

If bleeding persists, specialist referral is indicated. The nose will usually be packed with 1 cm ribbon gauze impregnated with bismuth iodoform petroleum paste (BIPP) and left for 48 hours. Post-nasal packing may be required if the bleeding originates from the back of the nose. In the rare cases that bleeding persists despite these local measures, maxillary artery or anterior ethmoidal artery ligation may be indicated.

Nasal fractures

Fractures of the nose are usually caused by a direct blow. Symptoms and signs of nasal fracture include pain,

epistaxis, swelling, crepitus, deformity and mobility of the nose. Nasal distortion may not be obvious once soft tissue swelling develops. Initial management is directed towards controlling the nasal haemorrhage. An associated laceration should be sutured with 6/0 nylon and requires prophylactic antibiotic therapy. The nasal passages should be examined to exclude a septal haematoma and patients should be advised to return if they notice increased pain or develop a fever (see below).

Radiographs are probably not required as undisplaced fractures require no treatment and displaced fractures are clinically obvious. Displaced nasal fractures may require reduction. There are two indications for reduction of fractures. The first is obstruction of the nasal passages and the second is cosmetic deformity.

In young athletes, displaced fractures are almost always reduced because of a tendency towards increased sinus infections and a decrease in the size of the nasal passage. Attempts at immediate reduction of nasal fractures are associated with a risk of arterial damage and severe acute haemorrhage. It is preferable to delay fracture reduction and refer the patient to a surgeon within 7 days of the injury. When the soft tissue swelling has settled sufficiently, reduction, if necessary, can be carried out under general anaesthesia.

Septal haematoma

This important condition can complicate what seems to be a trivial nosebleed. A septal haematoma is caused by haemorrhage between the two layers of mucosa covering the septum (Fig. 22.3). The presenting complaint is either nasal obstruction or nasal pain. The patient may be febrile and nasal examination reveals a cherry-like structure (the dull, red swollen septum) that occludes the nasal passages. Treatment of a large septal haematoma involves evacuation of the clot using a wide-bore needle or through a small incision followed by nasal packing to prevent recurrence of the haematoma. Antibiotic prophylaxis should be given to prevent development of a septal abscess and subsequent cartilage necrosis.

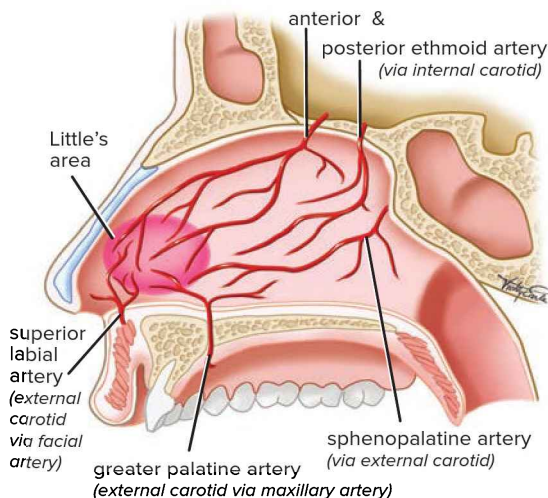


Figure 22.2 Little's (Kiesselbach's) area

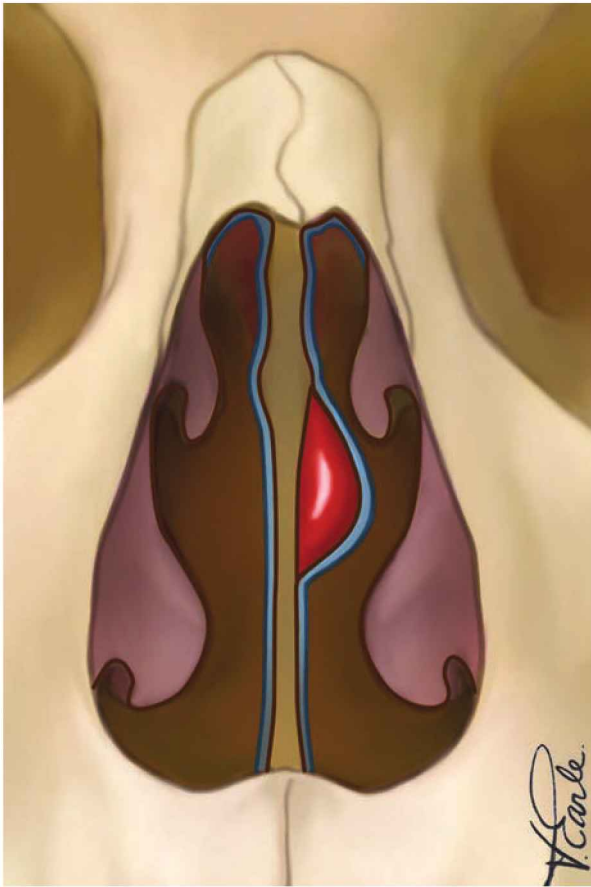


Figure 22.3 Septal haematoma

EAR

Ear injuries in most sports are not common. Nonetheless, they can pose a clinical challenge to sports physicians working in sports such as rugby and wrestling. The most frequent injury is a contusion to the ear known as an auricular haematoma.

Auricular haematoma

This injury occurs mainly in rugby scrums, boxing or wrestling as a result of a shearing blow. Recurrent contusions result in haemorrhage between the perichondrium and the cartilage. This may eventually develop into a chronic swelling, commonly known as 'cauliflower ear'. An acute haematoma (Fig. 22.4) should be treated initially with ice and firm compression, but may need to be drained by aspiration under strict aseptic conditions. A pressure dressing (cotton wool soaked in collodion) is then applied and is carefully packed against the ear to follow the contours of the outer ear. This is bandaged firmly. The ear must be examined daily to assess progress. Return to



Figure 22.4 Acute auricular haematoma

non-contact sports can be immediate, but headgear or a helmet is required for return to contact sport. Rugby forwards frequently wear headgear which protects their ears as a preventive measure.

Lacerations to the ear require careful cleansing and suture. As lacerations located between the scalp and the ear are easily missed, this area should always be examined, especially if there is a history of the ear being pulled forwards. If the auricular cartilage is torn, the overlying skin should be carefully aligned and sutured with absorbable 5/0 sutures; however, the cartilage itself has natural physical 'memory' properties, leading to natural alignment without the need for sutures. The perichondrium should be closed as a separate layer. Prophylactic oral antibiotic therapy is recommended.

Perforated eardrum

A blow across the side of the head may occasionally injure the eardrum. Pain, bleeding from the ear or impaired hearing suggest tympanic membrane rupture.

These ruptures usually heal spontaneously. Prophylactic antibiotic therapy (amoxycillin 250-500 mg, 8 hourly if not allergic to penicillin) should be administered. It is important to keep the ear dry while a perforation is present. In sports where significant pressure changes occur, such as platform diving, scuba diving and high-altitude mountain climbing, athletes should not return to play until the tympanic membrane has healed. Athletes participating in water sports, such as swimming and water polo, should use custom-fabricated ear plugs to maintain a dry ear canal. Dry land athletes may return to play as soon as any vertigo has resolved.¹

Athletes will often ask whether it is safe to fly with a perforated eardrum. It is generally considered safe to do so, unless they have had surgery to repair the perforation (myringoplasty), in which case guidance should be sought from the surgeon.

A severe blow across the head may fracture the skull and cause inner ear bleeding. Discharge from the ear (otorrhoea) may signal a neurosurgical emergency and, thus, patients should be referred immediately for specialist treatment.

Otitis externa

Otitis externa is the most common ear condition affecting competitive swimmers. It is generally caused by bacteria, although fungal infection can also contribute. Symptoms include earache, pruritus, discharge and impaired hearing. On examination, there may be discharge in the ear and local redness along the external auditory meatus. There may be tragal tenderness and pain on tragal pull.

Management involves careful aural toilet combined with topical antibiotic and corticosteroid ear drops. The patient should, preferably, abstain from swimming until fully recovered and avoid rubbing or drying the ear until after the infection has cleared. The use of earplugs in this condition is controversial. They may traumatise the ear canal and predispose the swimmer to infection.

PRACTICE PEARL

Recurrent attacks of otitis externa may be prevented by instillation of alcohol ear drops, for example, 5% acetic acid in isopropyl alcohol (aqua ear®), after each swimming session.

EYES

Eye injuries occur most commonly in stick sports, racquet sports, especially squash, and contact sports.² All eye injuries, even those that appear to be minor, require thorough examination. All serious eye injuries should be referred immediately to an ophthalmologist. The indications for immediate referral to an ophthalmologist are shown in the box below.

Athletes with a previous history of impaired vision in one or both eyes or previous eye trauma or surgery should be evaluated by an ophthalmologist prior to participating in a high-risk sport.

Assessment of the injured eye

To assess the injured eye, it is important to understand the anatomy. The anatomy of the eye is shown in Figure 22.5.

For a thorough assessment of the injured eye, an 'eye injuries kit' (Fig. 22.6) is very useful and can be carried as part of the 'physician's bag' (Chapter 47).

History

The history of the eye injury provides useful diagnostic information. Seek to discover the history of the mechanism of injury. Were glasses, contact lenses or a protective device being worn at the time of injury? Note any previous eye injury or problems. Ask about symptoms such as pain, blurred vision, loss of vision, flashing lights and diplopia.

Examination

Test the visual acuity of each eye using a Snellen chart, with and without glasses or contact lenses. If a Snellen chart is not available, use pages of a newspaper with

Clinical indications for immediate referral to a specialist ophthalmologist

Symptoms

Severe eye pain
Persistent blurred or double vision
Persistent photophobia

Signs

Suspected penetrating injury (corneal laceration, pear-shaped pupil)
Hyphaema
Embedded foreign body
No view of fundus (suspected vitreous haemorrhage or retinal detachment)
Markedly impaired visual acuity: 6/12 or less
Loss of part of visual field

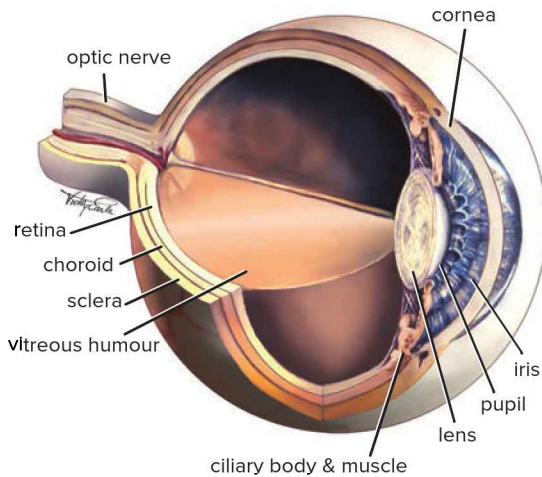


Figure 22.5 Anatomy of the eye



Figure 22.6 Eye injuries kit

variable print sizes as an approximate assessment of visual acuity. On the sporting arena, a scoreboard can be used to test distant vision.

Eye injuries kit

Small mirror
Pencil torch
Ophthalmoscope
Snellen chart
Sterile solution for irrigation
Local anaesthetic eye drops
Fluorescein
Antibiotic drops and ointments
Cotton buds
Contact lens lubricant and case(s)
Eye patches
Tape

Inspect the eyelids for bruising, swelling or laceration. Note any obvious foreign body, haemorrhage or change in pupil size.

If there is pain or photophobia due to the injury prevent examination of the eye, instil a drop of a sterile topical anaesthetic agent, such as amethocaine, to assist examination.

PRACTICE PEARL

Inspect the cornea for foreign material and abrasions. Fluorescein staining under cobalt blue light will help reveal areas of corneal ulceration or foreign bodies. Evert the upper lid to exclude the presence of a subtarsal foreign body.

Test eye movements in all directions. A restriction in any direction or the presence of diplopia may indicate orbital fracture. Compare the size, shape and light reaction of the pupil with the uninjured eye. An enlarged, poorly reacting pupil may be present after injury to the iris. A pear-shaped pupil suggests the presence of a full thickness corneal or scleral laceration (penetrating injury).

Inspect the anterior chamber for the presence of blood. Blood in the anterior chamber is known as hyphaema (Fig. 22.7).

PRACTICE PEARL

Ophthalmoscopic examination should routinely be performed to inspect the lens, vitreous humour and retina.

The absence of a red reflex on ophthalmoscopic examination may be due to a corneal opacity, a lens opacity (cataract), intraocular bleeding or a retinal detachment. Failure to visualise the fundus may be a sign of vitreous



Figure 22.7 Hyphaema—note the fluid level in the anterior chamber

haemorrhage, which can result from a retinal tear. Contusion of the retina may produce retinal oedema, seen as areas of pallor (and thickening) as well as retinal haemorrhage.

PRACTICE PEARL

Radiological examination of the orbit is indicated in all cases of traumatic eye injury with diplopia and in cases where an intraocular or intraorbital foreign body is suspected.

Corneal injuries: abrasions and foreign body

Corneal injuries in sport include abrasion, foreign body and, less commonly, alkali burn. Corneal abrasion, one of the most frequent injuries to the eye during sport, occurs as a result of a scratch from either a fingernail or foreign body. The patient complains of pain, a sensation of a foreign body being present in the eye and blurred vision if the central cornea is involved.

A topical anaesthetic drop should be instilled to assist corneal examination. Fluorescein staining under cobalt blue light will help locate corneal abrasions or foreign bodies. Evert the upper lid to exclude a subtarsal foreign body.

Treatment of corneal abrasions includes the instillation of antibiotic eye drops, for example, chloramphenicol and padding of the eye (Fig. 22.8). If pain and photophobia are severe, add a topical mydriatic (e.g. 2% homatropine). A local anaesthetic agent should never be used for pain relief as it can delay healing and result in further damage.

Eye-injury 'game-day' assessment

1. Test the visual acuity of each eye using a Snellen chart, with and without glasses or contact lenses (use newspaper pages or a scoreboard if unavailable)
2. Inspect the eyelids for bruising, swelling or laceration. Note any obvious foreign body, haemorrhage or change in pupil size (instil a drop of sterile topical anaesthetic agent to assist examination if pain or photophobia prevent examination)
3. Evert the upper lid to exclude the presence of a subtarsal foreign body
4. Inspect the cornea for foreign material and abrasions (fluorescein staining under cobalt blue light will assist greatly)
5. Test eye movements in all directions
6. Compare the size, shape and light reaction of the pupil with the uninjured eye
7. Inspect the anterior chamber for the presence of blood
8. Ophthalmoscopic examination should be performed to inspect the lens, vitreous humour and retina.

Corneal foreign bodies can be removed with a cotton tip applicator by a trained clinician. If foreign bodies are more deeply embedded, patients should be referred to an ophthalmologist. Rust rings, which occasionally remain after metallic foreign bodies have been embedded in the cornea, require removal by an ophthalmologist. Antibiotic eye ointment should be administered following foreign body removal and the eye padded for 24 hours until the corneal epithelium has healed.

If an athlete has sustained an alkali burn (from line markings), irrigate the eye copiously for 20 minutes with sterile saline or tap water and instil a local anaesthetic agent to assist this. The athlete should be seen as soon as possible by an ophthalmologist.

Subconjunctival haemorrhage

Trauma to the conjunctiva may cause subconjunctival haemorrhage—a bright red area in the white conjunctiva. Unless the haemorrhage is extensive or visual symptoms or photophobia are present, it is not clinically important.

PRACTICE PEARL

Subconjunctival haemorrhage is a classic finding with periorbital fractures, which need to be ruled out if the athlete participates in a sport where facial trauma is possible.



Figure 22.8 Eye padding

Blood pressure should be measured to rule out hypertension. Subconjunctival haemorrhage may obscure a perforation of the globe. If this is suspected, the patient should be referred to an ophthalmologist. In most cases, however, the athlete merely requires reassurance.

Eyelid injuries

Direct trauma to the eyelids may cause a large amount of bruising, which should be treated with cold compresses in the first 24 hours. Haemorrhage may spread subcutaneously across the midline to the other eye. A coexisting orbital fracture needs to be excluded in these patients.

Lacerations of the eyelid require meticulous primary repair. Each anatomical layer (conjunctiva, tarsal plate and skin) should be repaired separately by an ophthalmic or plastic surgeon.

Trauma near the medial canthus may lacerate the upper or lower lacrimal canaliculus (tear duct). If this is not repaired, the patient may have permanent watering of the eye. Such injuries require ophthalmological referral for microscopic suturing of the cut ends of the canaliculus.

PRACTICE PEARL

In all injuries to the eyelids, the eye also needs to be examined to exclude ocular injury.

Hyphaema

Bleeding into the anterior chamber of the eye results from ruptured iris vessels and may only be visible with slit lamp examination. More significant bleeds will present with a bloody layer in the anterior chamber (Fig. 22.7). In hyphaemas of small volume, visual acuity may be unaffected. Associated injuries may occur and all patients with a hyphaema should be referred to an ophthalmologist.

The aim of treatment of this condition is to prevent further bleeding, which may, in turn, result in uncontrollable glaucoma or blood staining of the cornea. The patient needs to rest in bed while the haemorrhage clears, usually over 3–5 days. Aspirin and other anti-inflammatory medications should be avoided as these may provoke further bleeding.

Lens dislocation

Blunt trauma may result in varying degrees of lens displacement. Partial dislocation causes few symptoms. Complete lens dislocation results in blurred vision. A common sign of lens dislocation is a quivering of the iris when the patient moves the eye. Iritis and glaucoma are possible sequelae of lens dislocation. Immediate ophthalmological referral is required. Surgical removal of the displaced lens may be indicated.

Vitreous haemorrhage

Bleeding into the vitreous humour signifies damage to the retina, choroid or ciliary body. Ophthalmoscopic examination reveals loss of the red reflex and a hazy appearance. Treatment generally consists of bed rest, but more severe cases may require specialist referral for removal of the blood and vitreous humour.

Retinal haemorrhage

Injury to the retina can result from a direct blow to the eye or a blow to the back of the head. Valsalva manoeuvres (e.g. in weightlifting) may also produce retinal oedema and haemorrhage. The patient may remain asymptomatic if peripheral areas of the retina are affected. Central retinal damage, however, blurs vision. On ophthalmoscopic examination, central retinal oedema appears as a white opacity that partially obscures the retinal vessels. Boxers may develop atrophic macular holes and loss of central vision as a result of recurrent contusive injuries. This condition requires specialist management.

Retinal detachment

Retinal detachment is more common in those with extreme myopia and may result from any blunt or perforating trauma. The patient complains of flashes of light or the appearance of a 'curtain' spreading across the field of vision, which can present months or even years after the initial injury. Ophthalmoscopic examination reveals elevation and folding of the detached retina, which trembles with each eye movement. Immediate referral for surgical treatment is indicated. An unusual case of retinal detachment in sport occurred in a swimmer who received an accidental blow to the goggles.³

Prevention of eye injuries

Athletes with certain eye problems should avoid contact sports altogether. These problems include:

- functionally only one eye
- severe myopia
- Marfan syndrome
- previous retinal detachment.

For squash, protective eyewear must be worn by people who have either only one good eye, amblyopia (lazy eye), recent eye surgery, history of pre-retinal detachment conditions or diabetic retinopathy. Protective eyewear should meet the Australian Standard AS4066 1992 or the US Standard ASTM F803.

Contact lenses offer no protection against eye trauma. Hard contact lenses are not suitable for sporting activity and should never be used in contact sport. Soft lenses appear to be reasonably safe in contact sport. One of the most common 'crises' in injury management is a lost contact lens. The athlete will complain that the contact lens is no longer in its correct position and cannot be located. Usually the lens has been displaced and with careful examination can be located elsewhere on the eye, often at the lower lid. Occasionally, the lens is displaced completely from the eye and lost on the playing surface. Those who wear contact lenses during sport should always carry a spare pair of contact lenses or a pair of protective spectacles as a back-up.

Those athletes who cannot or do not like to wear contact lenses can use protective goggles made of polycarbonate, which are available for most prescriptions. These polycarbonate goggles are also used as eye protection in sports with a high risk for eye injuries. The most obvious examples of these are squash and racquetball, where the size of the ball enables it to enter the orbit and compress the globe. The routine use of closed goggles is strongly recommended.

Certain sports require protection not only of the eye, but of the other facial structures. In sports such as American football, ice hockey, cricket and lacrosse, protective helmets and faceguards should also provide adequate eye protection. Because of the profound effect of major eye injury, we encourage athletes and sport governing

bodies to be proactive in promoting and enforcing the use of eye protection where indicated.⁴

TEETH

Trauma to the teeth and dental tissues is a relatively common occurrence in certain sports. A large study, from over 70 000 episodes of basketball, showed mouth guard wearers had a reduced number of dental injuries and referrals to the dentist when compared to those who did not wear mouth guards.⁵ This illustrates the importance of the team doctor or coach in preventing injuries through education of athletes and also the requirement to know how to manage dental trauma, when it occurs.

Nature of injuries

Within the mouth are a number of tissues which can be traumatised as seen in the image below (Fig. 22.9).

Trauma results in tissues being crushed, stretched or torn, as seen in the image of a dento-alveolar fracture (Fig. 22.10). The damage caused to teeth and their supporting tissues following trauma is dependent on a number of factors, such as the energy of the impact, the direction of the blow, the tissues traumatised, the resilience or elasticity of the striking object, and the local anatomy. Therefore, dental trauma may present in a number of different ways despite apparently similar causes.

Emergency management

Options for treatment of dental injuries at a sporting event may be limited, due to lack of equipment and facilities. Some injuries may not be visible and may require further special investigations to diagnose. However, many injuries do not need immediate treatment and the most common injuries seen are detailed in Table 22.3 below.

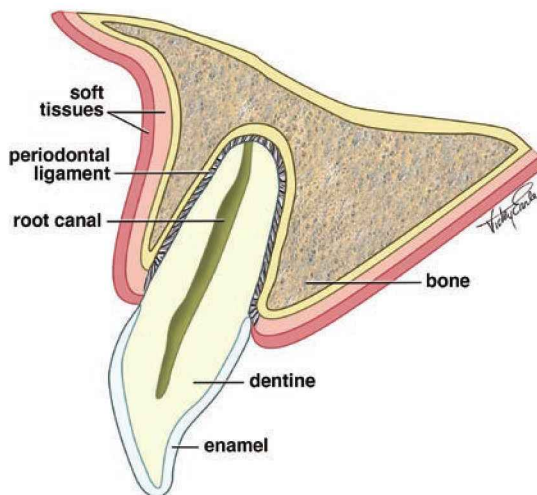


Figure 22.9 Sagittal view showing the jaw and a tooth in the maxillary incisor region identifying the tissues that may be damaged following dental trauma

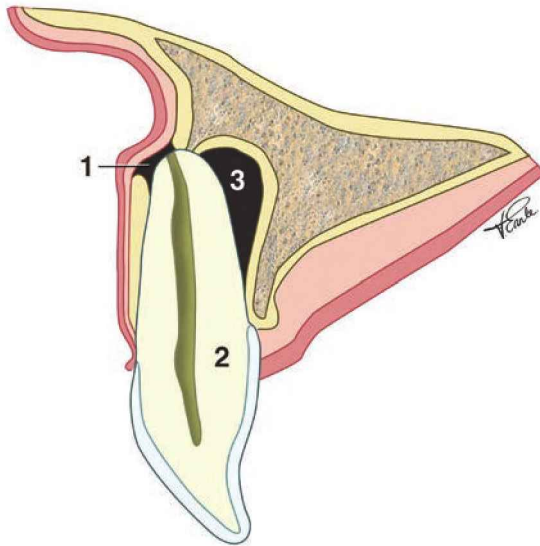


Figure 22.10 Dento-alveolar fracture showing fracture (1) and displacement of a tooth (2) from its socket (3)

Certain situations require special consideration in dental trauma:

- Mouth guard in situ—in general, leave a mouth guard in place as removal might further damage traumatised teeth. If mouth guard must be removed, ensure the patient is seated upright and check the mouth and mouth guard following its removal for teeth and teeth fragments. The mouth guard should not be used to splint teeth due to the problems of oral hygiene.
- Dentures, crowns and bridges—if loose, then remove if possible as they present a choking hazard.
- Unconscious patients—store avulsed teeth in a suitable storage medium and pass to the Emergency Department.

Dental management and follow-up

Injured athletes must be examined by a dentist as soon as possible, with a referral letter if time permits. Treatment

can be complex and complications of healing are common. Therefore, athletes who experience trauma should be informed they need close follow-up at regular intervals over a number of years.

Prevention

Although prevention of injuries is often thought to just include the wearing of protective mouth or headguards, an important aspect is education. Athletes should be individually assessed to determine if guards should be worn, for example when engaging in high-risk contact sports, and if prominent upper incisor teeth are present. All efforts should be made to encourage their use.

The two most common types of protection are listed below.

Faceguards:

- are pre-formed cages that attach to a helmet
- provide protection from blows to the front of the head
- often fail to prevent similar blows to the chin, therefore mouth guards also may need to be worn
- significantly reduce injuries in certain sports such as hockey⁶ and baseball.⁷

Mouth guards:

- use has also reduced the frequency of dental trauma-related injuries⁸
- prevent damage to teeth, soft tissues and mandible, by cushioning against the opposing teeth following a direct blow
- can also prevent teeth from being lost from the mouth, swallowed or inhaled
- do not appear to prevent concussion injuries.⁹

Three types of mouth guard exist (Fig. 22.11):

1. Stock mouth guard (Fig 22.11a)—loose fitting, made of plastic or rubber and should be avoided as they provide none of the protective functions already discussed.

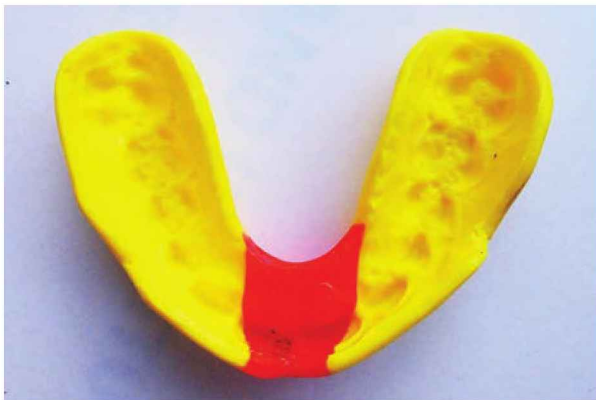


(a)



(b)

Figure 22.11 (a) Stock mouth guard (b) Mouth-formed mouth guards



(c)

Figure 22.11 (cont.) (c) Custom-made mouth guard**Figure 22.12** Maxillary incisors and canines showing how tooth/root axes are inclined towards the back of the mouth to guide tooth replantation if avulsed**Table 22.3** Emergency dental treatment

Injury	Treatment required
Fractured teeth	• Collect tooth fragments and store in milk or saline • Fragments can be re-attached to teeth in the dental surgery • Assume lost fragments have been swallowed or inhaled. These must be located with chest or abdominal radiographs as the consequences of inhaling infected tooth fragments are serious • Attached fragments should be left unless they can be easily detached
Avulsed tooth	• Reposition in socket as soon as possible (within 5 minutes¹⁰) • Hold avulsed tooth up by its crown, not root • Wash with saline or water for 10 seconds only, until the root looks clean • Re-implant the tooth in its socket • Check it is correctly aligned and the athlete can bite together • Bite on clean gauze • Visit general doctor for a tetanus booster if needed
Multiple avulsed teeth	• Teeth are unique in shape and their sockets closely replicate their contour • Gently insert into their sockets, one at a time • If teeth do not fit in their sockets, try another socket • Do not force teeth into their sockets • Upper central incisors are larger than lateral incisors • Lower incisors are difficult to tell apart • As a guide, the root tips of teeth are tilted towards the back of the mouth (Fig 22.12)
Displaced teeth	• Gently reposition • If any resistance to repositioning or pain are felt, the athlete should attend their dentist for this to be performed under local anaesthetic
Avulsed teeth that cannot be reimplanted	• Teeth must be kept in a suitable storage medium for transport to the dentist. Never store in water • Type of storage medium depends on storage time: – <10 minutes—physiological saline – Up to several hours¹¹—fresh milk (cold, low fat is best)¹² – Up to 24 hours¹³—special culture medium, available commercially as Dentosafe® or EMT™ Tooth Saver • If no medium is available, store in athlete's saliva

2. Mouth-formed mouth guards (Fig. 22.11b)—two types:
 - a. a shell filled with a plastic or rubber to improve fit
 - b. pre-formed plastic mouth guard softened in hot water and moulded to the shape of the teeth and soft tissues

Both types can be tight fitting, but can be bulky, less comfortable and therefore less likely to be used.
3. Custom made (Fig. 22.11c)—constructed on accurate models of the athlete's teeth. They are well retained and less bulky than other designs, but are more expensive to purchase. Due to their size and fit, they are more comfortable to wear than other types of mouth guards, and as a result more likely to be used in sport.

Mouth guards should be kept clean using standard mouthwashes to rinse. They should be kept away from extreme heat, such as hot water, which will cause distortion of fit.

Therefore a custom-made mouth guard from a dentist is advised, as this will be the most comfortable, least bulky and best fitting. If not available, consider using one that can be moulded in the mouth.

FRACTURES OF FACIAL BONES

In sport, facial fractures may result from blows by implements such as bats or sticks, equipment such as skis¹⁴ and from collision injuries. Mountain biking is a sport that causes a significant proportion of facial injuries. Eye wear¹⁵ and face shields¹⁶ can protect against facial injuries.

Symptoms and signs range from pain, swelling, ipsilateral epistaxis, laceration and bruising to gross deformity. Examination may reveal facial asymmetry, crepitus, discoloration or obvious deformity. Most facial fractures are not overly painful and so there should be little objection to firm palpation, which aids in the diagnostic process. Objectively, the bite should be examined for malocclusion while the patient should also be subjectively asked whether their bite 'feels' out of alignment. Bimanual examination of the facial bones may show areas of discontinuity and mobility. If maxillary fracture is suspected, the upper teeth can be grasped to determine evidence of excessive movement of the upper jaw and midface. Opening and closing the jaw may reveal pain, limitation or deviation with mandibular injuries.

Initial management of facial fractures is directed towards maintenance of the patient's airway. In mandibular body fractures or maxillary fractures, this may require emergency manual reduction. Associated head and cervical spine injuries should be excluded. The oral cavity requires inspection for bleeding or dental damage. CT scans have supplanted plain radiographs as the imaging modality of choice.

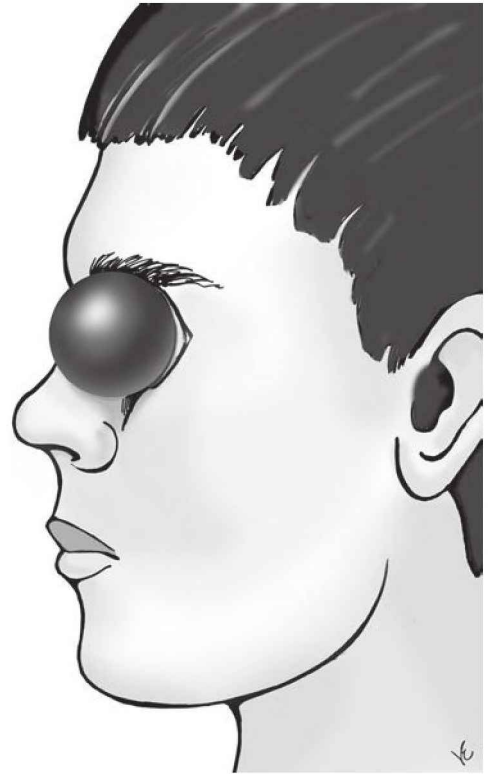


Figure 22.13 A squash ball fits precisely into the eye socket

Orbital fracture with and without globe trauma

'Blow-out' fracture of the orbit results from direct trauma such as a fist, cricket ball, baseball¹⁷ or squash ball (Fig. 22.13). Compression of the globe and orbital contents produces a fracture in the weakest part of the orbit, the orbital floor. Contents of the orbit may herniate through the defect. The patient typically presents with a periorbital haematoma, protruding or sunken eye, double vision on upward gaze and numbness of the cheek. Double vision on upward gaze is due to the entrapment of the inferior rectus muscle in the fracture.

A detailed examination of the eye must be performed to exclude intraocular injuries such as hyphaema, lens dislocation or ruptured globe. If an orbital fracture is suspected, a birds-eye plain radiograph should be performed. The radiograph may not show the fracture, but may demonstrate some clouding of the maxillary sinus. CT examination is used to confirm the fracture. Children require early surgery and as such should be investigated with high clinical suspicion.

Antibiotic therapy should be commenced immediately and the patient referred to an ophthalmologist. Surgery may be required to release the trapped muscle and repair

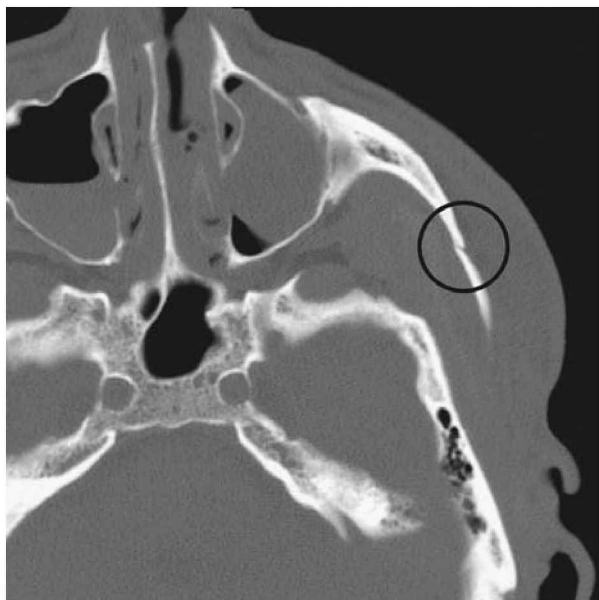


Figure 22.14 CT scan confirming a fractured zygomatic arch; plain radiography did not detect the fracture

the bony defect. Athletes who undergo surgery should be warned not to blow their nose, due to the risk of displacing the fracture and further haemorrhage.

Fractures of the zygomaticomaxillary complex

Zygomaticomaxillary complex fractures (Fig. 22.14) occur from a direct blow to the cheek such as from a fist, hockey stick or baseball. Signs include swelling and bruising, flatness of the cheek and mandibular function disturbance. If associated with an orbital fracture, there may be concomitant diplopia, limitation of ocular movement and asymmetry of the eyes, as well as numbness of the affected cheek, nose, upper lips and upper teeth.

Surgical treatment consists of closed or open reduction under general anaesthesia. Unstable fractures require

fixation. Associated orbital fractures are treated by open reduction and reconstruction of the orbital floor.

Maxillary fractures

Maxillary fractures usually result from a direct high-velocity crushing blow to the middle portion of the face, and are extremely rare in sports other than motor sports. They are classified as Le Fort I, II and III fractures (Fig. 22.15) depending on whether the nasal or cheek bones are involved. Le Fort I fractures result in the separation of the maxilla from the nasal-septal structures and the pterygoid plates. Clinically, Le Fort I fractures are identified when the entire maxilla moves as a separate unit. Le Fort II fractures separate the maxilla and the nasal complex from the orbital-zygomatic structures. On clinical examination the maxilla and nose move together as one unit. Le Fort III fractures separate the maxillary, zygomatic, nasal and orbital structures from the cranial base (Fig. 22.16).¹⁸

Maxillary fractures are often accompanied by cranial damage, obstruction of the nasal airway, oedema of the soft palate, haemorrhage into the sinuses and disturbance of the contents of the orbit. CSF rhinorrhoea may occur, indicating fracture of the cribriform plate. Reduced sensation in the infraorbital region is common.

Examination findings include lengthening of the face, midface mobility, malocclusion of the bite and periorbital bruising. Initial treatment is aimed at protecting the airway—the conscious patient should sit leaning forward. This should be followed by rapid transfer for definitive diagnosis and treatment. Surgical treatment involves reduction and fixation with plates and screws.

Mandibular fractures

Fracture of the mandible is one of the most common facial fractures in sport and usually results from a direct blow. The most common fracture sites are the mandibular angle and the condyle. The mandible usually breaks in more than one place as a result of the trauma and these

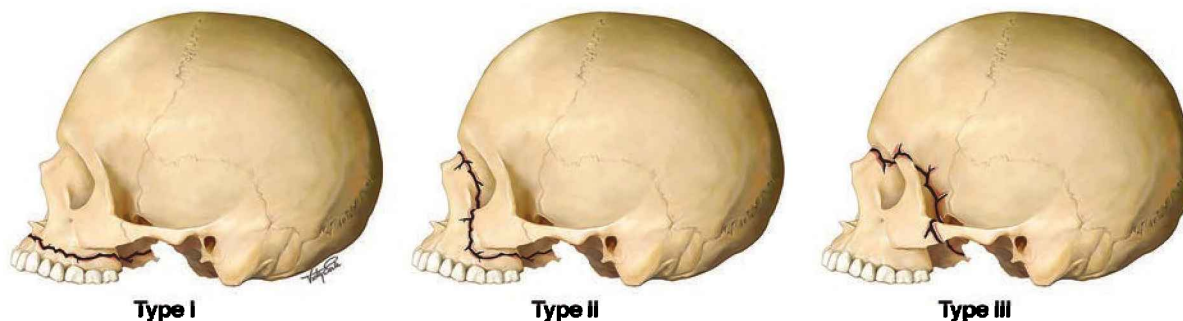


Figure 22.15 Le Fort I, II and III fractures



Figure 22.16 CT scan of Le Fort III fracture (arrow)

fractures usually occur on opposite sides of the midline. Fractures may be undisplaced or displaced.

Undisplaced fractures

Minor mandibular fractures are painful, tender and swollen. These are managed conservatively with analgesia and rest. The patient should eat soft food only for up to 4 weeks as symptoms resolve.

Displaced mandibular fractures

Displaced mandibular fractures are severe injuries that result from considerable force. Alveolar (tooth-bearing) fractures are the most common type. These fractures range from single tooth fractures or avulsions to complete segment mobility. The clinical diagnosis is obvious when two or more teeth move as a unit.

Inspection may reveal malalignment of teeth and bruising to the floor of the mouth. Palpation reveals malocclusion, tenderness and defects along the lower border of the mandible. Paraesthesia or anaesthesia of the lower lip and chin suggest damage to the inferior alveolar nerve.

Initial treatment includes maintenance of the airway in a forward sitting position with the patient's hands supporting the lower jaw. A jaw bandage can be used in comminuted or badly displaced fractures, although this is generally not used due to the increased associated risk of compromising the airway by causing backward displacement of the mandible. A cervical collar can be used as an alternative. A concussed or unconscious patient should be placed in a lateral position with head tilt and jaw

support after the mouth has been cleared of any dislodged teeth or tooth fragments. Occasionally, the tongue may need to be held forward to maintain an open airway.

Most displaced mandibular fractures require closed reduction and intermaxillary fixation for 4–6 weeks. If adequate closed reduction cannot be achieved, then open reduction and internal fixation is required. A fracture of one condyle usually does not require immobilisation except to control pain. Active jaw exercises should be commenced as soon as pain permits.

During the period of intermaxillary fixation, the athlete may perform mild exercises such as stationary bike riding and light weightlifting. Resumption of contact sport should be delayed until at least 1–2 months after the jaws are unwired. Earlier resumption is possible when internal fixation has been used. The use of a protective polycarbonate facial shield may offer some protection if early return to play is contemplated.

Patients with mandibular fractures who are eating soft food or have their jaws wired can be referred to a dietitian for advice on suitable liquid meals and foods suitable for vitamising.

Temporomandibular injuries

Blows to the mandible can produce a variety of temporomandibular joint (TMJ) injuries. Trauma to the jaw while the mouth is open occasionally produces TMJ dislocation. Other injuries include haemarthrosis, meniscal displacement and intracapsular fracture of the head of the condyle.

Examination of the injured TMJ may reveal limitation of opening, pain and malocclusion. Dislocation of the TMJ causes inability to close the mouth. A dislocated TMJ may be reduced by placing both thumbs along the line of the lower teeth as far posteriorly as possible and applying downward and backward pressure. Longstanding dislocations may require general anaesthesia for reduction. Management of TMJ dislocation includes rest with limitation of mouth opening for up to 7–10 days, a soft diet and analgesics such as aspirin. Contact sport should be avoided for up to 2 weeks depending on the symptoms. Boxers should not attempt sparring for at least 6 weeks.

Chronic temporomandibular pain

Chronic TMJ problems are sometimes referred to as 'temporomandibular joint dysfunction' or 'myofascial pain dysfunction syndrome'. This syndrome appears to affect males more than females with a peak incidence in the early twenties. Patients complain of pain, limitation of movement, clicking and locking of the TMJ. Treatment should include assessment by a dentist to exclude any

malocclusion problem. Physiotherapy evaluation with a view to manual therapy and exercise therapy can be invaluable.

PREVENTION OF FACIAL INJURIES

Protective equipment has been designed for sports where facial injury is a risk (Chapter 9). Properly designed helmets have reduced the incidence of faciomaxillary injuries. Helmets are designed for a single impact or multiple impacts, and should be individually fitted for

each athlete. Single-impact helmets such as most pushbike and motorbike helmets must be discarded after the user has had a fall.

Facial protective equipment is constantly developing, with items such as face shields, cages and eyeglasses often used in high-risk sports such as ice-hockey and baseball. Such equipment may also be a useful adjunct to minimise any re-injury risk in the return-to-play process. The legality of the equipment should always be checked with the sport's governing body.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e